Chapter 7. Inference in Multiple Regression I

Gauss–Markov, confidence intervals, general linear hypotheses, residuals

Contents

7.1 The Gauss–Markov theorem **2**

7.2 Confidence intervals **2**

7.3 The general linear hypothesis **3**

7.3.1 Method I: constrained least squares 3

7.3.2 Method II: full and reduced models 4

7.4 Properties of residuals **5**

7.5 Selected exercises **5**

Throughout, $\mathbf{y} = X\beta + \epsilon$ with X of size $n \times (p+1)$ and full column rank. We distinguish two sets of assumptions:

- **(A)** $\epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n)$, needed for exact t and F inference;
- **(B)** $E\epsilon = \mathbf{0}$ and $\operatorname{Var}\epsilon = \sigma^2 I_n$ only, enough for the optimality of least squares.

7.1 The Gauss–Markov theorem

Motivation. Why least squares? Among all linear unbiased estimators, it has the smallest variance for every linear combination of coefficients, under uncorrelated errors with equal variances and without normality.

Theorem 7.1 (Gauss–Markov)

Under (B), among all linear unbiased estimators $\tilde{\beta} = A\mathbf{y}$ of β, the least squares estimator $\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$ has the smallest variance: $\operatorname{Var}\tilde{\beta} - \operatorname{Var}\hat{\beta}$ is nonnegative definite. Equivalently, for every $\mathbf{c}$, $\mathbf{c}'\hat{\beta}$ has the smallest variance among linear unbiased estimators of $\mathbf{c}'\beta$.

Proof [Supplement]. $E(A\mathbf{y}) = AX\beta = \beta$ for all β forces $AX = I$. Write $A = (X'X)^{-1}X' + D$; then $DX = AX - I = 0$. Hence

$$\operatorname{Var}(A\mathbf{y}) = \sigma^2 AA' = \sigma^2\{(X'X)^{-1} + (X'X)^{-1}X'D' + DX(X'X)^{-1} + DD'\} = \sigma^2(X'X)^{-1} + \sigma^2 DD',$$

and DD' is nonnegative definite, with $\mathbf{c}'DD'\mathbf{c} = \|D'\mathbf{c}\|^2$. Equality for all $\mathbf{c}$ holds iff $D = 0$. ■

So $\hat{\beta}$ is the **best linear unbiased estimator** (BLUE). The theorem needs neither normality nor independence, only uncorrelated errors with a common variance. If $\operatorname{Var}\epsilon = \sigma^2 V$ with $V \neq I$, the BLUE is the generalized least squares estimator (Chapter 12).

7.2 Confidence intervals

Motivation. Individual coefficient intervals follow from the t distribution, but each one describes the effect of a predictor holding the others fixed, which is why coefficients change when variables are added.

Under (A), with $(X'X)^{-1} = (c_{ij})$ and $MSE = SSE/(n-p-1)$ independent of $\hat{\beta}$ (Theorem 6.4):

Theorem 7.2

$100(1-\alpha)\%$ confidence intervals are:

(a) for a coefficient: $\hat{\beta}_j \pm t_{\alpha/2}(n-p-1)\sqrt{c_{jj}MSE}$;

(b) for a linear combination $\mathbf{q}'\beta$: $\mathbf{q}'\hat{\beta} \pm t_{\alpha/2}(n-p-1)\sqrt{\mathbf{q}'(X'X)^{-1}\mathbf{q}\,MSE}$;

(c) for the mean response at $\mathbf{x}$: $\mathbf{x}'\hat{\beta} \pm t_{\alpha/2}(n-p-1)\sqrt{\mathbf{x}'(X'X)^{-1}\mathbf{x}\,MSE}$;

(d) a prediction interval for a new observation at $\mathbf{x}$: $\mathbf{x}'\hat{\beta} \pm t_{\alpha/2}(n-p-1)\sqrt{\{1 + \mathbf{x}'(X'X)^{-1}\mathbf{x}\}MSE}$.

With σ^2 known, replace MSE by σ^2 and $t_{\alpha/2}$ by $z_{\alpha/2}$.

Proof. (b) contains (a) and (c) as special cases. $\mathbf{q}'\hat{\beta} \sim N(\mathbf{q}'\beta, \sigma^2\mathbf{q}'(X'X)^{-1}\mathbf{q})$ is independent of $(n-p-1)MSE/\sigma^2 \sim \chi^2(n-p-1)$, so the studentized ratio is $t(n-p-1)$.

(d) adds the variance of the new error. ■

7.3 The general linear hypothesis

Many hypotheses are linear restrictions on β:

$$H_0 : C\beta = \mathbf{m} \quad \text{vs} \quad H_1 : C\beta \neq \mathbf{m},$$

with C a known $k \times (p+1)$ matrix of rank k and $\mathbf{m} \in \mathbb{R}^k$. Examples, for $E(y) = \beta_0 + \beta_1 x_1 + \cdots + \beta_4 x_4$:

- $\beta_1 = \beta_2$ and $\beta_3 = 2\beta_4$: $C = \begin{pmatrix} 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \end{pmatrix}$, $\mathbf{m} = \mathbf{0}$;
- $\beta_1 = \beta_2 = \beta_3 = \beta_4$: three rows $(0, 1, -1, 0, 0)$, $(0, 0, 1, -1, 0)$, $(0, 0, 0, 1, -1)$;
- $\beta_j = 0$: $C = \mathbf{e}_j'$, the j-th unit row.

There are two equivalent routes: minimize SSE under the restriction, or reparametrize to a reduced model.

7.3.1 Method I: constrained least squares

Theorem 7.3 (Restricted least squares)

The minimizer of $\|\mathbf{y} - X\beta\|^2$ subject to $C\beta = \mathbf{m}$ is

$$\tilde{\beta} = \hat{\beta} - (X'X)^{-1}C'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m}),$$

and its residual sum of squares is

$$SSE(\tilde{\beta}) = SSE + Q, \qquad Q = (C\hat{\beta} - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m}).$$

Proof [Supplement]. With a Lagrange multiplier vector $\lambda \in \mathbb{R}^k$, minimize $L = (\mathbf{y} - X\beta)'(\mathbf{y} - X\beta) + 2\lambda'(C\beta - \mathbf{m})$. Setting derivatives to zero:

$$X'X\tilde{\beta} + C'\lambda = X'\mathbf{y}, \qquad C\tilde{\beta} = \mathbf{m}.$$

The first gives $\tilde{\beta} = \hat{\beta} - (X'X)^{-1}C'\lambda$, and substituting into the second, $C\hat{\beta} - C(X'X)^{-1}C'\lambda = \mathbf{m}$, so $\lambda = \{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m})$. ($C(X'X)^{-1}C'$ is invertible since C has full row rank.) That this stationary point is the constrained minimum follows from the decomposition below. For any β,

$$\|\mathbf{y} - X\beta\|^2 = \|\mathbf{y} - X\hat{\beta}\|^2 + (\hat{\beta} - \beta)'X'X(\hat{\beta} - \beta),$$

since $X'(\mathbf{y} - X\hat{\beta}) = \mathbf{0}$. With $\beta = \tilde{\beta}$ and $\hat{\beta} - \tilde{\beta} = (X'X)^{-1}C'\lambda$,

$$(\hat{\beta} - \tilde{\beta})'X'X(\hat{\beta} - \tilde{\beta}) = \lambda'C(X'X)^{-1}C'\lambda = (C\hat{\beta} - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\hat{\beta} - \mathbf{m}) = Q.$$

For any feasible β, $(\hat{\beta} - \beta)'X'X(\hat{\beta} - \beta) \geq Q$ (it is the squared distance, in the $X'X$ metric, from $\hat{\beta}$ to the affine set $\{C\beta = \mathbf{m}\}$, attained at $\tilde{\beta}$), so $\tilde{\beta}$ is optimal. ■

Theorem 7.4 (F test of the general linear hypothesis)

Under (A),

$$\frac{Q}{\sigma^2} \sim \chi^2(k, \lambda), \qquad \lambda = \frac{(C\beta - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\beta - \mathbf{m})}{2\sigma^2},$$

Q is independent of SSE, and

$$F_0 = \frac{Q/k}{SSE/(n-p-1)} = \frac{Q}{k \cdot MSE} \sim F(k, n-p-1, \lambda).$$

Under H_0, $\lambda = 0$, and the test rejects when $F_0 > F_\alpha(k, n-p-1)$.

Proof. $C\hat{\beta} - \mathbf{m} \sim N(C\beta - \mathbf{m}, \sigma^2 C(X'X)^{-1}C')$, and Q is the corresponding quadratic form with matrix $W = \{C(X'X)^{-1}C'\}^{-1}$. Since $W \cdot \sigma^2 C(X'X)^{-1}C'/\sigma^2 = I_k$ is idempotent, Theorem 5.5 gives the noncentral chi-square with k degrees of freedom. Q is a function of $\hat{\beta}$, which is independent of SSE (Theorem 6.4). ■

7.3.2 Method II: full and reduced models

Substitute the restriction into the model to get a **reduced model** with $p + 1 - k$ free parameters; fit it and compute

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/k}{SSE(F)/(n-p-1)},$$

where $SSE(F) = SSE$ is from the full model, with $df_F = n - p - 1$ and $df_R = n - p - 1 + k$. If the restriction involves a nonzero $\mathbf{m}$, move the known part to the left-hand side (an offset). Since $SSE(R)$ is exactly the constrained minimum $SSE(\tilde{\beta})$, Theorem 7.3 gives $SSE(R) - SSE(F) = Q$: *the two methods give the same statistic.*

For a model without intercept, $\mathbf{y} = X\beta + \epsilon$ with X of size $n \times p$, the same holds with $n - p$ in place of $n - p - 1$.

Example 7.1

A no-intercept model $y = \beta_1 x_1 + \beta_2 x_2 + \epsilon$ is fitted to

x_1	1	2	0	1	3
x_2	0	1	1	2	1
y	3	7	2	5	10

Test $H_0 : \beta_1 = 2\beta_2$. Here

$$X'X = \begin{pmatrix} 15 & 7 \\ 7 & 7 \end{pmatrix}, \qquad X'\mathbf{y} = \begin{pmatrix} 52 \\ 29 \end{pmatrix}, \qquad \hat{\beta} = \frac{1}{56}\begin{pmatrix} 7 & -7 \\ -7 & 15 \end{pmatrix}\begin{pmatrix} 52 \\ 29 \end{pmatrix} = \begin{pmatrix} 2.875 \\ 1.268 \end{pmatrix},$$

$\mathbf{y}'\mathbf{y} = 187$, and $SSE = 187 - \hat{\beta}'X'\mathbf{y} = 0.732$ with 3 degrees of freedom, so $MSE = 0.244$. *Method I*: $C = (1, -2)$, $C\hat{\beta} = 0.339$, $C(X'X)^{-1}C' = \frac{7+28+60}{56} = 1.696$, so $Q = 0.339^2/1.696 = 0.068$ and $F_0 = 0.068/0.244 = 0.28$. *Method II*: under H_0, $y = \beta_2(2x_1 + x_2) + \epsilon$; with $z = 2x_1 + x_2 = (2, 5, 1, 4, 7)$, $\sum z^2 = 95$ and $\sum zy = 133$, so $SSE(R) = 187 - 133^2/95 = 0.800$ and $\{SSE(R) - SSE\}/MSE = 0.068/0.244 = 0.28$. Both give $F_0 = 0.28 < F_{0.05}(1, 3) = 10.13$: H_0 is not rejected.

7.4 Properties of residuals

Motivation. Residuals are not the errors: they are correlated and have unequal variances $\sigma^2(1-h_{ii})$. This motivates the standardized and studentized residuals of Chapter 10.

Theorem 7.5

For the fitted model with intercept:

(a) $\sum_i e_i = 0$, $\sum_i x_{ij} e_i = 0$ for each j, and $\sum_i \hat{y}_i e_i = 0$;

(b) under (A), $\mathbf{e} = (I - H)\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2(I - H))$; in particular $\operatorname{Var} e_i = \sigma^2(1 - h_{ii})$ and $\operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$.

Proof. (a) is $X'\mathbf{e} = \mathbf{0}$ together with $\hat{\mathbf{y}}'\mathbf{e} = \hat{\beta}' X'\mathbf{e}$.

(b) $E\mathbf{e} = (I - H)X\beta = \mathbf{0}$ and $\operatorname{Var}\mathbf{e} = \sigma^2(I - H)(I - H)' = \sigma^2(I - H)$. ■

So residuals are correlated, with correlation $-h_{ij}/\sqrt{(1 - h_{ii})(1 - h_{jj})}$, and have unequal variances depending on the design. Plots of residuals against $\hat{y}_i$, against each x_{ij}, and against time order are the basic checks:

- a horizontal band: no problem;
- a funnel: nonconstant variance; use weighted least squares or transform y;
- a linear trend against x_j: a computational error, since (a) forces zero correlation;
- a curve: add a quadratic term (or other function) of x_j, or transform.

7.5 Selected exercises

Motivation. The general linear hypothesis is applied to a model whose mean is itself constrained and to weights that must sum to one; a last exercise shows what imposing a restriction does to the bias and the variance of the estimator.

Exercise 7.1 (An F statistic under a side condition)

Let $\mathbf{y} = \theta + \epsilon$ with $\mathbf{y}, \theta, \epsilon \in \mathbb{R}^4$ and $\epsilon \sim N_4(\mathbf{0}, \sigma^2 I_4)$, where θ is known to satisfy $\theta_1 + \theta_2 + \theta_3 + \theta_4 = 0$. Show that the F statistic for $H_0 : \theta_1 = \theta_3$ is

$$F_0 = \frac{2(y_1 - y_3)^2}{(y_1 + y_2 + y_3 + y_4)^2},$$

and state its null distribution.

Solution.

Step 1. (The model as a regression.) The full model says $\theta \in \Omega = \{\theta : \mathbf{1}'\theta = 0\}$, a 3-dimensional subspace. With the basis $(1, -1, 0, 0)'$, $(0, 1, -1, 0)'$, $(0, 0, 1, -1)'$ of Ω as columns of a 4×3 matrix X of full rank, the model is $\mathbf{y} = X\beta + \epsilon$ without intercept, and H_0 is one linear restriction on β. So Method II of Section 7.3 applies: $F_0 = \{SSE(R) - SSE(F)\}/\{SSE(F)/(4-3)\}$, with 1 numerator degree of freedom. For a model whose mean ranges over a subspace L, $SSE = \|\mathbf{y} - P_L\mathbf{y}\|^2$ with P_L the orthogonal projection onto L (Theorem 6.1).

Step 2. (Full model.) Ω is the orthogonal complement of $\mathbf{1}$, so $P_\Omega \mathbf{y} = \mathbf{y} - \bar{y}\mathbf{1}$ and

$$SSE(F) = \|\bar{y}\mathbf{1}\|^2 = 4\bar{y}^2 = \frac{(y_1 + y_2 + y_3 + y_4)^2}{4}.$$

Step 3. (Reduced model.) Let $\mathbf{c} = (1, 0, -1, 0)'$. Then $\mathbf{c} \in \Omega$, and the reduced mean space is $\omega = \{\theta \in \Omega : \mathbf{c}'\theta = 0\}$, the orthogonal complement of $\mathbf{c}$ inside Ω. Hence $\Omega = \omega \oplus \operatorname{span}(\mathbf{c})$ orthogonally and $P_\Omega = P_\omega + P_{\mathbf{c}}$, with $P_{\mathbf{c}}\mathbf{y} = (\mathbf{c}'\mathbf{y}/\|\mathbf{c}\|^2)\mathbf{c}$. Since $\mathbf{y} - P_\Omega\mathbf{y} \perp \Omega \ni P_{\mathbf{c}}\mathbf{y}$, Pythagoras gives

$$SSE(R) = \|\mathbf{y} - P_\Omega\mathbf{y} + P_{\mathbf{c}}\mathbf{y}\|^2 = SSE(F) + \frac{(\mathbf{c}'\mathbf{y})^2}{\|\mathbf{c}\|^2} = SSE(F) + \frac{(y_1 - y_3)^2}{2}.$$

Step 4. Therefore

$$F_0 = \frac{(y_1 - y_3)^2/2}{(y_1 + y_2 + y_3 + y_4)^2/4} = \frac{2(y_1 - y_3)^2}{(y_1 + y_2 + y_3 + y_4)^2},$$

and under H_0, $F_0 \sim F(1, 1)$ by Theorem 7.4 (with $n - p$ in place of $n - p - 1$ for a model without intercept, here $4 - 3 = 1$). ■

Exercise 7.2 (Weights that sum to one)

In a survey, n respondents report scores $X_{i1}, \ldots, X_{ip}$ on p items and an overall score Y_i. The model is

$$Y_i = \sum_{j=1}^{p} \omega_j X_{ij} + \epsilon_i, \qquad \sum_{j=1}^{p} \omega_j = 1, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n),$$

with $X = (X_{ij})$ of size $n \times p$ and full rank, $n > p$ (negative weights allowed). Derive the least squares estimator of $\omega = (\omega_1, \ldots, \omega_p)'$ and an unbiased estimator of σ^2, and give their distributions.

Solution. Write $G = (X'X)^{-1}$, $\hat{\omega} = GX'\mathbf{Y}$ (the unconstrained estimator), $\mathbf{g} = G\mathbf{1}$ and $s = \mathbf{1}'G\mathbf{1} > 0$.

Step 1. (Estimator.) Theorem 7.3 with $C = \mathbf{1}'$ ($k = 1$) and $\mathbf{m} = 1$ gives

$$\tilde{\omega} = \hat{\omega} - \frac{\mathbf{1}'\hat{\omega} - 1}{s}\mathbf{g} = \Big(I - \frac{\mathbf{g}\mathbf{1}'}{s}\Big)\hat{\omega} + \frac{\mathbf{g}}{s},$$

and indeed $\mathbf{1}'\tilde{\omega} = \mathbf{1}'\hat{\omega} - (\mathbf{1}'\hat{\omega} - 1) = 1$.

Step 2. (Reduced model.) Substituting $\omega_p = 1 - \sum_{j<p} \omega_j$,

$$Y_i - X_{ip} = \sum_{j<p} \omega_j (X_{ij} - X_{ip}) + \epsilon_i,$$

an unconstrained regression of $\mathbf{Y} - \mathbf{X}_p$ on $Z = XB$, where B is the $p \times (p-1)$ matrix with columns $\mathbf{e}_j - \mathbf{e}_p$. B has rank $p - 1$ and X full column rank, so Z has full column rank $p - 1$. The map $(\omega_1, \ldots, \omega_{p-1}) \mapsto \omega$ is a bijection onto the constraint set, and $\|\mathbf{Y} - X\omega\|^2$ equals the residual sum of squares of the reduced regression at $(\omega_1, \ldots, \omega_{p-1})$. Hence $\tilde{\omega}$ is this regression's estimator (completed by $\tilde{\omega}_p = 1 - \sum_{j<p} \tilde{\omega}_j$), and $SSE(\tilde{\omega}) = \|\mathbf{Y} - X\tilde{\omega}\|^2$ is its residual sum of squares, with $n - (p - 1)$ degrees of freedom.

Step 3. (σ^2.) Theorem 6.4, applied to the reduced regression (its proof uses only that the hat matrix of Z is the projection onto the column space of Z), gives $SSE(\tilde{\omega})/\sigma^2 \sim \chi^2(n - p + 1)$, independent of the reduced coefficients and hence of $\tilde{\omega}$. Thus

$$\tilde{\sigma}^2 = \frac{\|\mathbf{Y} - X\tilde{\omega}\|^2}{n - p + 1}$$

is unbiased, and $(n - p + 1)\tilde{\sigma}^2/\sigma^2 \sim \chi^2(n - p + 1)$.

Step 4. (Distribution of $\tilde{\omega}$.) $\tilde{\omega}$ is an affine function of $\hat{\omega} \sim N_p(\omega, \sigma^2 G)$, hence normal. Since $\mathbf{1}'\omega = 1$, $E\tilde{\omega} = \omega - \frac{1-1}{s}\mathbf{g} = \omega$. With $M = I - \mathbf{g}\mathbf{1}'/s$,

$$\operatorname{Var}\tilde{\omega} = \sigma^2 MGM' = \sigma^2\Big(G - \frac{\mathbf{g}\mathbf{g}'}{s} - \frac{\mathbf{g}\mathbf{g}'}{s} + \frac{\mathbf{g}(\mathbf{1}'G\mathbf{1})\mathbf{g}'}{s^2}\Big) = \sigma^2\Big(G - \frac{\mathbf{g}\mathbf{g}'}{s}\Big).$$

So $\tilde{\omega} \sim N_p(\omega, \sigma^2(G - \mathbf{g}\mathbf{g}'/s))$, a singular normal distribution concentrated on the hyperplane $\mathbf{1}'\omega = 1$: indeed $(G - \mathbf{g}\mathbf{g}'/s)\mathbf{1} = \mathbf{g} - \mathbf{g} = \mathbf{0}$. Replacing σ^2 by $\tilde{\sigma}^2$ gives t intervals for each ω_j with $n - p + 1$ degrees of freedom. ■

Exercise 7.3 (Bias and variance of restricted least squares)

Under assumption (B), let $\tilde{\beta}$ be the restricted estimator of Theorem 7.3, and $G = (X'X)^{-1}$, $W = \{CGC'\}^{-1}$.
(a) Show that $E\tilde{\beta} = \beta - GC'W(C\beta - \mathbf{m})$; in particular $\tilde{\beta}$ is unbiased when the restriction is true.
(b) Show that $\operatorname{Var}\tilde{\beta} = \sigma^2(G - GC'WCG)$, so that $\operatorname{Var}\hat{\beta} - \operatorname{Var}\tilde{\beta}$ is nonnegative definite whether or not the restriction is true.

Solution [Supplement].

(a) By Theorem 7.3, $\tilde{\beta} = M\hat{\beta} + GC'W\mathbf{m}$ with $M = I - GC'WC$. Since $E\hat{\beta} = \beta$,

$$E\tilde{\beta} = \beta - GC'WC\beta + GC'W\mathbf{m} = \beta - GC'W(C\beta - \mathbf{m}),$$

which equals β when $C\beta = \mathbf{m}$.

(b) $\operatorname{Var}\tilde{\beta} = M(\sigma^2 G)M'$. Using $G' = G$, $W' = W$ and $W(CGC')W = W$,

$$\begin{aligned} MGM' &= (G - GC'WCG)(I - C'WCG) \\ &= G - GC'WCG - GC'WCG + GC'W(CGC')WCG = G - GC'WCG. \end{aligned}$$

Hence $\operatorname{Var}\hat{\beta} - \operatorname{Var}\tilde{\beta} = \sigma^2(CG)'W(CG)$, and $\mathbf{a}'(CG)'W(CG)\mathbf{a} = \mathbf{b}'W\mathbf{b} \geq 0$ with $\mathbf{b} = CG\mathbf{a}$, since W is positive definite. So a correct restriction yields an unbiased estimator with smaller variance, while a false one buys the same variance reduction at the price of the bias in (a). ■